Fiscal Year:	FY 2008	**Task Last Updated:**	FY 08/25/2008
PI Name:	Platts, Steven H. Ph.D.		
Project Title:	Determination of the Magnitude and Time Course of Cardiovascular Alterations During a Simulated Extended Stay Lunar Mission		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	281-483-8177
Organization Name:	NASA Johnson Space Center		
PI Address 1:	Cardiovascular Laboratory		
PI Address 2:	Biomedical Research and Environmental Sciences Division		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Crew Health NNJ07ZSA002N
Start Date:	07/01/2008	**End Date:**	06/30/2010
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Goodwin, Thomas	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 6/30/2010 (from 6/30/2011) per PI (Ed., 4/4/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Martin, David (Wyle Labs) Stenger, Michael (Wyle Labs)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Cardiovascular adaptations to microgravity exposure, such as decreased plasma volume, cardiac function and orthostatic tolerance are well documented. It is unknown if extended habitation in a partial Earth gravity environment, such as that on the Moon, will be protective against these effects of spaceflight deconditioning. Computer modeling predicts that extended duration missions to the Moon will result in 6% plasma volume loss, which is approximately half that seen in spaceflight. The effects of this level of hypovolemia on cardiac function and orthostatic tolerance are not clear. We propose to study the time course of the fluid shifts and plasma volume losses experienced during a simulated trip to, and extended habitation on, the Moon. Additionally, we propose to evaluate the effect of this volume change on cardiac function using newer preload-independent echocardiographic techniques. By acquiring daily measurements during the transitional phases of changing gravitational levels, we plan to validate the current computational model and further quantify the crew health and performance risks associated with extended duration missions to the Moon and other partial Earth gravity environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New project for FY2008.
Bibliography Type:	Description: (Last Updated: 03/01/2018)